Spatial Data Quality Assessment Report

This report presents the evaluation of spatial data quality in the study area of Balzan, Malta. The analysis focuses on assessing horizontal and vertical positional accuracy, as well as evaluating metadata completeness, attribute accuracy, logical consistency, and temporal accuracy.

High-quality spatial data is essential for reliable decision-making. Therefore, this study aims to identify inconsistencies, quantify positional errors, and assess the overall reliability of the datasets.

Methodology

The positional accuracy assessment was conducted by comparing a test dataset with an independent reference dataset of higher accuracy.

For horizontal accuracy, differences in X and Y coordinates were calculated for each point. For vertical accuracy, elevation differences (ΔZ) were computed.

The Root Mean Square Error (RMSE) was used to quantify positional accuracy. Accuracy values were then derived at the 95% confidence level.

Outlier detection was performed using a rejection threshold based on error magnitude. Values exceeding this threshold were considered outliers and excluded from calculations to ensure a reliable representation of dataset quality.

Horizontal Accuracy

The horizontal dataset demonstrates a high level of consistency, with most positional differences remaining very small.

The calculated RMSE is **0.2038 m**, resulting in a horizontal accuracy of **0.3528 m** at the 95% confidence level.

A rejection level of **0.611** was applied to identify extreme outliers. One point was identified as an outlier due to a significantly large positional error (approximately 64), which clearly deviates from the rest of the dataset.

After removing this outlier, the dataset shows a stable distribution of errors. The low variability and small standard deviation indicate that positional errors are tightly clustered, confirming the high reliability of the horizontal data.

Vertical Accuracy

The vertical dataset shows significantly higher variability compared to the horizontal dataset.

The computed RMSE is **5.706 m**, corresponding to a vertical accuracy of **11.184 m** at the 95% confidence level.

Several points were identified as outliers due to large deviations:

- $\Delta Z \approx 12.3$ m
- $\Delta Z \approx 18.3$ m
- $\Delta Z \approx 25.0$ m

These values are considerably higher than the rest of the dataset and clearly represent anomalies.

After excluding these outliers, the dataset becomes more consistent; however, the vertical accuracy remains lower than horizontal accuracy.

This difference reflects the inherent limitations of elevation data, which is affected by interpolation processes, terrain complexity, and resolution constraints of Digital Terrain Models.

Outlier Analysis

Outliers represent extreme deviations that can distort statistical results and lead to misleading conclusions.

In the horizontal dataset, only **1 out of 40 points (2.5%)** was removed, indicating high consistency and reliability.

In contrast, **3 out of 31 points (9.68%)** were removed in the vertical dataset, reflecting greater variability and lower reliability of elevation data.

The removal of outliers ensured that the computed RMSE values provide a realistic representation of positional accuracy.

Metadata Assessment

The metadata of the datasets (`Map_Points_INDEPENDENT` and `GCPoints_TEST`) was reviewed and updated using GIS software.

Several key metadata elements were added and completed, including:

- Title
- Summary (Description)
- Tags
- Credits
- Use limitations
- Lineage

These additions improve dataset documentation, enhance usability, and ensure compliance with standard metadata requirements.

Proper metadata is essential for understanding dataset origin, processing methods, and usage constraints. The lack of complete metadata can reduce data transparency and limit its applicability in future analyses.

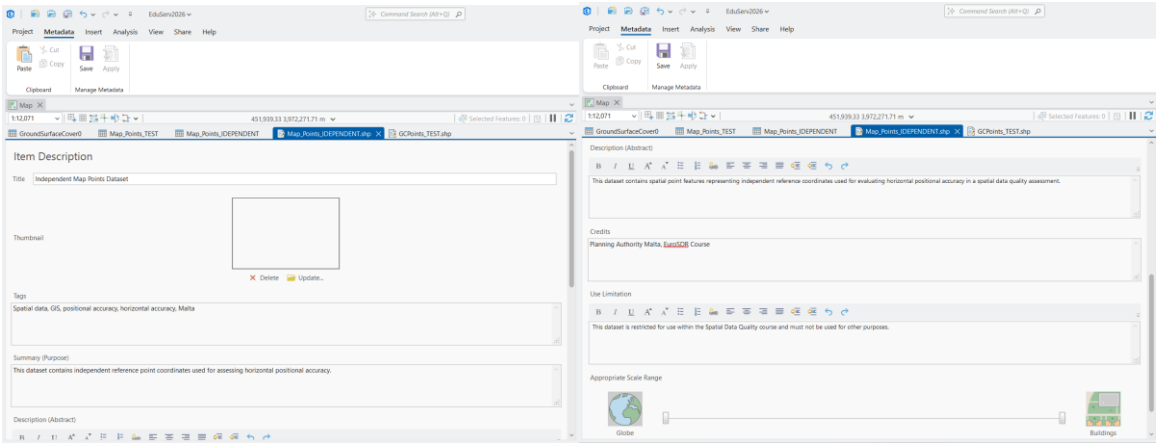

Figure 1: Updated metadata elements for the Map_Points_INDEPENDENT dataset.

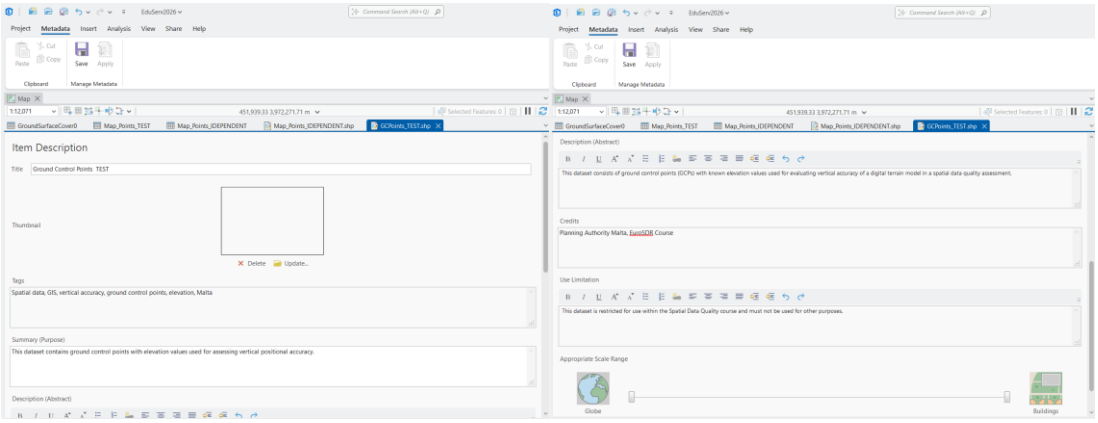

Figure 2: Updated metadata elements for the GCPoints_TEST dataset.

The updated metadata includes essential elements such as title, description, tags, and lineage, improving data documentation and usability.

Attribute Accuracy

A visual comparison between the spatial dataset (`GroundSurfaceCover0`) and aerial imagery was performed to assess attribute accuracy.

Several inconsistencies were identified:

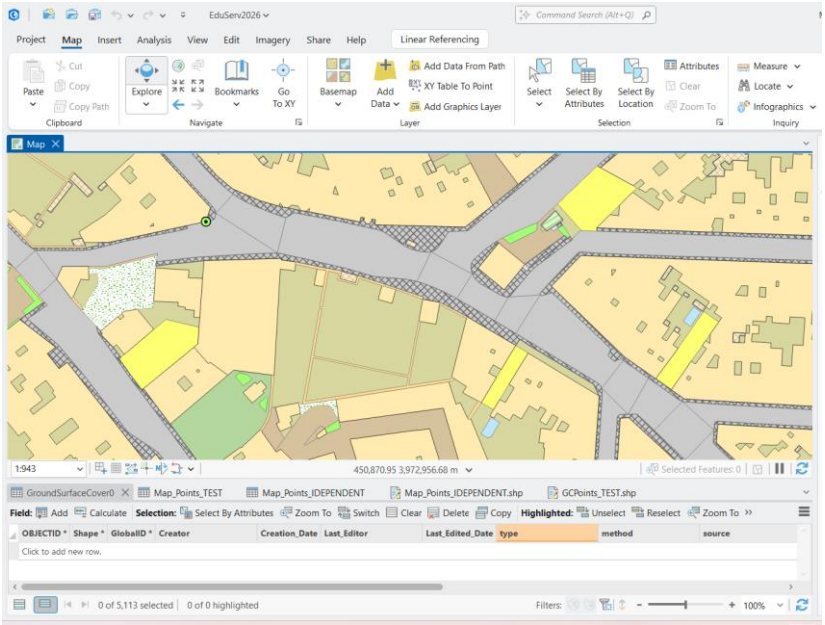

Figure 3: A construction site feature classified in the dataset appears to be a fully developed building in aerial imagery.

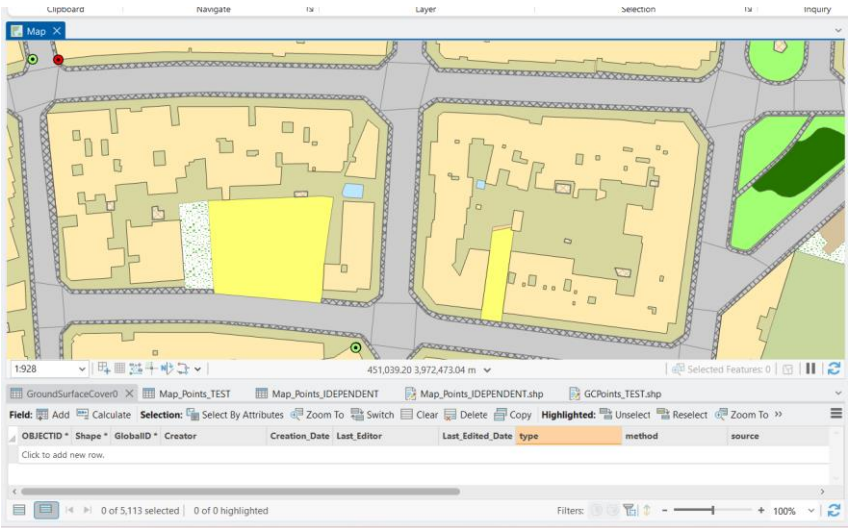

Figure 4: Construction site features located within built-up areas, indicating outdated or incorrect classification.

Features classified as **construction sites (yellow polygons)** were found within fully developed urban areas where buildings appear complete. These features do not exhibit characteristics of active construction, indicating misclassification and suggesting that the dataset is outdated.

Additionally, multiple construction site features located along streets and within built-up blocks do not represent actual construction activity. Similarly, some areas classified as **natural land (dotted pattern)** appear to be modified or partially developed, further indicating inconsistencies in attribute classification.

These findings demonstrate that attribute errors exist and highlight the importance of continuous data validation and updates. These inconsistencies indicate that the dataset is not fully up to date, highlighting issues related to both attribute and temporal accuracy.

Logical Consistency

Logical consistency refers to the internal structure of spatial data, ensuring that there are no gaps, overlaps, or invalid geometries.

Although a full topology validation was not performed, visual inspection did not reveal major inconsistencies. This suggests that the dataset maintains an acceptable level of structural integrity. No significant overlaps, gaps, or geometry errors were observed during visual inspection.

Temporal Accuracy

Temporal accuracy refers to how up-to-date spatial data is, and it plays a critical role in decision-making processes.

One example of temporal inaccuracy is flood risk management. If outdated elevation or land-use data is used, flood models may fail to identify newly developed areas, leading to underestimation of flood risks and potential damage.

Another example is urban planning. Using outdated spatial data may result in ignoring newly constructed buildings, roads, or land-use changes, leading to inefficient planning decisions and conflicts in land use. These examples highlight the importance of maintaining up-to-date spatial datasets to ensure accurate analysis and reliable decision-making.

Conclusion

The results indicate that the horizontal dataset provides high positional accuracy (**0.35 m**) and is suitable for high-precision applications.

The vertical dataset, with an accuracy of approximately 11.18 m, is less precise but still acceptable for general use.

Outlier detection played a crucial role in obtaining reliable results. The analysis also highlighted the importance of metadata completeness, attribute validation, and temporal accuracy in ensuring high-quality spatial data.

Overall, the dataset is of good quality, with the main limitations related to vertical accuracy and outdated attribute classifications.